



Federal Board HSSC-II Examination Physics Model Question Paper

(National Curriculum of Pakistan 2022-2023)
(Scheme of Studies 2006)

Section - A (Marks 17)

Time Allowed: 25 minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

ROLL NUMBER					

0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

Version No.			

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2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
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8	8	8	8
9	9	9	9

Candidate Sign. _____

Invigilator Sign. _____

Q1. Fill the relevant bubble against each question according to curriculum. Each part carries one mark.

Sr #	Question	(A)	(B)	(C)	(D)	(A)	(B)	(C)	(D)
1.	The distance between the two identical balls is R. If the distance is reduced to half of R then the gravitational force will become:	half	quarter	double	four time	☐	☐	☐	☐
2.	As we go further away from the surface of the Earth, the value of 'g':	remains same	increases	decreases	may increase or decrease depending on mass.	☐	☐	☐	☐
3.	SI unit of Luminosity is	watts	cd/m ²	joules	cd.m ²	☐	☐	☐	☐
4.	Real gases deviate from ideal behavior because gas molecules:	are colorless	are spherical	have intermolecular forces of attraction	have high speeds	☐	☐	☐	☐
5.	If A is the amplitude of an oscillating system, the total distance covered during its time period is:	A/2	A	2A	4A	☐	☐	☐	☐
6.	The phenomenon of diffraction is more prominent when the size of the obstacle is:	larger than the wavelength of incident light	smaller than the wavelength of incident light	comparable to the wavelength of incident light	independent of the wavelength of incident light	☐	☐	☐	☐
7.	Which of the following is equivalent to 1.0 volt.	newton per second	joule per second	joule per coulomb	coulomb per joule	☐	☐	☐	☐
8.	Capacitor is a device used to:	store electrical energy	vary the resistance	store magnetic energy	dissipate energy	☐	☐	☐	☐
9.	In a full wave bridge rectifier, how many diode(s) conduct during each half cycle of AC input?	1	2	3	4	☐	☐	☐	☐
10.	Sinusoidal waveform is generated in AC system because:	it can be easily drawn	it produces least disturbance	it is natural standard	other waves cannot be	☐	☐	☐	☐

$$V = \frac{W}{q}$$



			in electrical circuits		produced easily.	
11.	A photoelectric cell is a device which:	converts electricity into light energy.	converts light energy into electricity.	stores light energy.	stores electricity	☐ ☐ ☐ ☐
12.	If uncertainty in the position of an electron is zero, the uncertainty in its momentum will be	less than $h/4$	greater than $h/4$	zero	infinite	☐ ☐ ☐ ☐
13.	The process by which a heavy nucleus splits into two smaller nuclei is called:	Fission	Fusion	Alpha decay	Beta decay	☐ ☐ ☐ ☐
14.	Which law gives the relationship between brightness and distance from the source	Law of brightness	Direct Square Law of brightness	Inverse Square Law of brightness	Inverse Square Law of light	☐ ☐ ☒ ☐ $I \propto \frac{1}{r^2}$
15.	Which of the following statements is true about a photon?	A photon has zero mass and zero momentum.	A photon has finite mass and a finite value of momentum.	A photon has zero mass but finite value of momentum.	A photon has finite mass but zero momentum	☐ ☐ ☐ ☐
16.	The drawback of conducting a medical imaging test is:	Early detection of problem.	Accurate diagnosis.	Contribution to choose of effective treatment.	Exposure to radiation.	☐ ☐ ☐ ☐
17.	Energy of gamma photon having a wavelength of $1\text{\AA}$ is:	$12.4 \times 10^3 \text{ eV}$	$12.4 \times 10^4 \text{ eV}$	$12.4 \times 10^{-3} \text{ eV}$	$12.4 \times 10^{-4} \text{ eV}$	☐ ☐ ☐ ☐



PHYSICS HSSC-II

SECTION – A (Marks 17)

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حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ٹائم مرکز کے حوالے کر دیا جائے گا۔
لکھنے کی اجازت نہیں ہے۔ سیاہی خنجر کا استعمال ممنوع ہے۔

Version No.				
4	0	0	4	3

ROLL NUMBER						

①	●	●	○	○
①	①	①	①	①
②	②	②	②	②
③	③	③	③	●
●	④	④	●	④
⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨

○	○	○	○	○	○	○
①	①	①	①	①	①	①
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④	④	④	④	④	④	④
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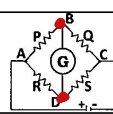
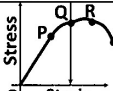
Answer Sheet No. _____

ہر سوال کے سامنے دیے گئے، کرکیم کے مطابق درست دائرہ کو پر کریں۔

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. Which of the following CANNOT be accelerated using electric field?	Proton	Ion	Electron	Neutron	☐	☐	☐	☐
2. According to Quantum theory, increasing the brightness of a beam of light without changing the color will increase:	The speed of photons	The frequency of photons	The number of photons	The energy of each photon	☐	☐	☐	☐
3. Primary function of transformer is to change:	The direction of alternating current	The level of alternating voltage	Alternating current into direct current	Mechanical energy into heat energy	☐	☐	☐	☐
4. A forward biased PN Junction offers:	Very high resistance	Very low resistance	Infinite resistance	Zero resistance	☐	☐	☐	☐
5. Which is NOT the characteristic of LASER?	Intense	Multi-directional	Monochromatic	Coherent	☐	☐	☐	☐
6. The resulting nucleus in given reaction is: ${}^1_0n + {}^{137}_{56}\text{Ba} \rightarrow \text{_____} + \gamma$	${}^{136}_{56}\text{Ba}$	${}^{138}_{57}\text{La}$	${}^{138}_{56}\text{Ba}$	${}^{138}_{55}\text{Cs}$	☐	☐	☐	☐
7. Compton effect proves the:	Particle nature of radiations	Wave nature of particle	Wave nature of radiations	Dual nature of particle	☐	☐	☐	☐
8. Which is an application of Mutual Induction?	AC generator	Transformer	Radio choke	DC generator	☐	☐	☐	☐
9. The temperature at which a ferromagnetic material becomes paramagnetic is called:	Neutral Point	Absolute Point	Inflection Point	Curie Point	☐	☐	☐	☐
10. SI unit of capacitive reactance is:	Ohm	Mho	Farad	Henry	☐	☐	☐	☐
11. In which of the following conditions, the galvanometer shows zero deflection?	 $V_B = V_D$	$V_B < V_D$	$V_B > V_D$	$V_B = 2V_D$	☐	☐	☐	☐
12. Due to electric polarization of dielectric, the charge storing capacity of capacitor:	Remains the same	Becomes zero	Decreases	Increases	☐	☐	☐	☐
13. In the given stress-strain curve which point is known as yield point?	 R	S	P	Q	☐	☐	☐	☐
14. Which of the following series lies in the ultraviolet region?	Bracket series	Pfund series	Lyman series	Balmer series	☐	☐	☐	☐
15. Which one belongs to Hadrons?	Photon	Pion	Electron	Neutrino	☐	☐	☐	☐
16. Coulomb force between two electric charges is 120N. If the distance between the charges is doubled, the coulomb force will be:	15 N	30 N	60 N	40 N	☐	☐	☐	☐
17. A particle having the mass of an electron and charge of proton is called:	Antiproton	Electron	Photon	Positron	☐	☐	☐	☐

—2HA-I 25004 (B)—

$$\Delta K.E = q\Delta V \quad \frac{1}{\lambda} = R_H \left[\frac{1}{p^2} - \frac{1}{n^2} \right] \quad R_H = 1.0974 \times 10^7 \text{ m}^{-1} \quad \varepsilon = \frac{\Delta \phi}{\Delta t} \quad X_c = \frac{1}{2\pi fC} \quad F = \frac{Kq_1q_2}{r^2}$$

$$F \propto \frac{1}{r^2} = 120 \times \frac{1}{4} = 30 \text{ N}$$



PHYSICS HSSC-II

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لیکھنے کی اجازت نہیں ہے۔ ایڈز/خوش کال استعمال نہ کریں۔

Version No.				
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ROLL NUMBER					

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Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

$$E = \frac{F}{q}$$

$$E = \frac{V}{d}$$

$$C = A\epsilon_0$$

$$\epsilon = V_t + Ir$$

$$V_t = \epsilon - Ir$$

Question	A	B	C	D	A	B	C	D
1. Which one belongs to Lepton's group?	Electron	Proton	Neutron	Pion	☐	☐	☐	☐
2. The unit of electric field intensity $\vec{E}$ is:	Vm	NC^{-2}	Vm^{-1}	NmA^{-1}	☐	☐	☐	☐
3. Capacitance of a capacitor can be increased by:	Using battery of high potential	Increasing distance between the plates	Decreasing area of plates	Inserting dielectric between the plates	☐	☐	☐	☐
4. Use Kirchhoff's first law to deduce the value of Current I in figure:	4A	1A	2A	3A	☐	☐	☐	☐
5. The correct relationship between ϵ and V_t for a closed circuit is:	$\epsilon = V_t - Ir$	$\epsilon = V_t$	$\epsilon > V_t$	$\epsilon < V_t$	☐	☐	☐	☐
6. Ohm-meter is the SI unit of:	Resistivity	Resistance	Conductivity	Conductance	☐	☐	☐	☐
7. Lenz's law is in accordance with the law of conservation of:	Mass	Charge	Momentum	Energy	☐	☐	☐	☐
8. Primary function of transformer is to change:	The level of alternating voltage	Alternating current into direct current	Mechanical energy into heat energy	The direction of alternating current	☐	☐	☐	☐
9. Magnetic flux will be minimum if the angle between magnetic field strength and area vector is:	90°	0°	45°	60°	☐	☐	☐	☐
10. At high frequency in an inductive circuit, the current will be:	Infinite	Large	Small	Zero	☐	☐	☐	☐
11. Identify the diamagnetic material:	Copper	Aluminium	Cobalt	Iron	☐	☐	☐	☐
12. In photoelectric effect, which factor increases by increasing the intensity of incident photon?	Stopping potential	Work function	K.E of electrons	Number of photons	☐	☐	☐	☐
13. The conversion of an alternating current into direct current is called:	Rectification	Amplification	Resolution	Magnification	☐	☐	☐	☐
14. An electron in a hydrogen atom is in its ground state. The minimum energy required to ionize it is:	1.5 ev	13.6 ev	3.4 ev	0.1 ev	☐	☐	☐	☐
15. Initial quantity of a uranium sample is 400g. After 3 rd half life how much uranium will be left?	25 g	50 g	100 g	200 g	☐	☐	☐	☐
16. Positron and electron annihilate into:	One γ -ray photon	α -rays	Two γ -ray photons	X-rays	☐	☐	☐	☐
17. Peak value of an AC voltage is 300 V, its rms value is:	212V	300V	220V	320V	☐	☐	☐	☐

—2HA-I 25004 (D)—

$\sum I = 0$ • $\Delta KE = q\Delta V$ • For $t = nT_1, N = \left(\frac{1}{2}\right)^n \times N_0$

$V_m = \sqrt{2}V_{rms}$ • $X_L = 2\pi fL$ • $E_n = \frac{E_0}{n^2}, E_0 = 13.6\text{ev}$

$V_m = \frac{300}{\sqrt{2}}$

$\epsilon = \frac{\Delta\phi}{\Delta t}$ • $\phi = \vec{B} \cdot \vec{A}$ • $E = \frac{F}{q} = \frac{\Delta V}{\Delta r}$

$44 \times \frac{1}{8} = 5.5$



PHYSICS HSSC-II

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Version No.				
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ROLL NUMBER							

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Answer Sheet No. _____

ہر سوال کے سامنے دیے گئے ہر یکولم کے مطابق درست دائرہ کو پر کریں۔

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. Which nuclear reaction is the source of energy in stars?	Chain reaction	Radioactivity	Fusion	Fission	☐	☐	☐	☐
2. The energy released in nuclear reactions is given by:	$Q = mc\Delta T$	$E = \Delta mc^2$	$F = ma$	$V = IR$	☐	☐	☐	☐
3. The electric dipole moment is the product of:	Charge and force	Charge and electric field	Charge and voltage	Charge and displacement	☐	☐	☐	☐
4. The SI unit of electric flux is:	V/m	C/N	N/C	$N \cdot m^2/C$	☐	☐	☐	☐
5. According to Gauss's law, the electric field inside a hollow conductor is:	Decreasing	Zero	Constant	Increasing	☐	☐	☐	☐
6. Which physical law relates current through a conductor to voltage and resistance?	Ampere's law	Faraday's law	Kirchhoff's law	Ohm's law	☐	☐	☐	☐
7. The force on a current carrying conductor placed in a magnetic field is maximum when the conductor is:	At 45° to the field	Out of the field	Perpendicular to the field	Parallel to the field	☐	☐	☐	☐
8. According to Faraday's law, the induced emf is proportional to the:	Rate of change of magnetic flux	Inverse of magnetic flux	Square of magnetic flux	Magnetic flux	☐	☐	☐	☐
9. In an AC circuit, the voltage and current are out of phase by 90° when the circuit is purely:	Inductive	Conductive	Resistive	Capacitive	☐	☐	☐	☐
10. The energy gap in conductors is:	Zero	Moderate	Very large	Small	☐	☐	☐	☐
11. The temperature at which a ferromagnetic material becomes paramagnetic is called:	Absolute temperature	Knee temperature	Critical temperature	Curie temperature	☐	☐	☐	☐
12. In forward biasing of a p-n junction diode, the potential barrier:	Decreases	Disappears	Increases	Remains constant	☐	☐	☐	☐
13. The photoelectric effect provides evidence for:	Dual nature of light	Diffraction of light	Particle nature of light	Wave nature of light	☐	☐	☐	☐
14. According to Einstein's equation of photoelectric effect, the maximum kinetic energy of emitted electrons depends upon:	Number of photons	Velocity of light	Frequency of light	Intensity of light	☐	☐	☐	☐
15. Conversion of A.C into D.C is called:	Oscillation	Rectification	Modulation	Amplification	☐	☐	☐	☐
16. Which of the following series of hydrogen spectra lies in the visible region of the spectrum?	Paschen series	Brackett series	Lyman series	Balmer series	☐	☐	☐	☐
17. The time required for half of a radioactive sample to decay is called:	Half-life	Decay constant	Mean life	Average life	☐	☐	☐	☐

—2HA-II 25004—

• $\phi_E = \vec{E} \cdot \vec{A}$ • $\vec{F} = I(\vec{L} \times \vec{B})$ • $\frac{1}{\lambda} = R_H \left[\frac{1}{p^2} - \frac{1}{n^2} \right]$ • $P = |Qd|$ • $V = IR$ • $\epsilon = \frac{\Delta \phi}{\Delta t}$